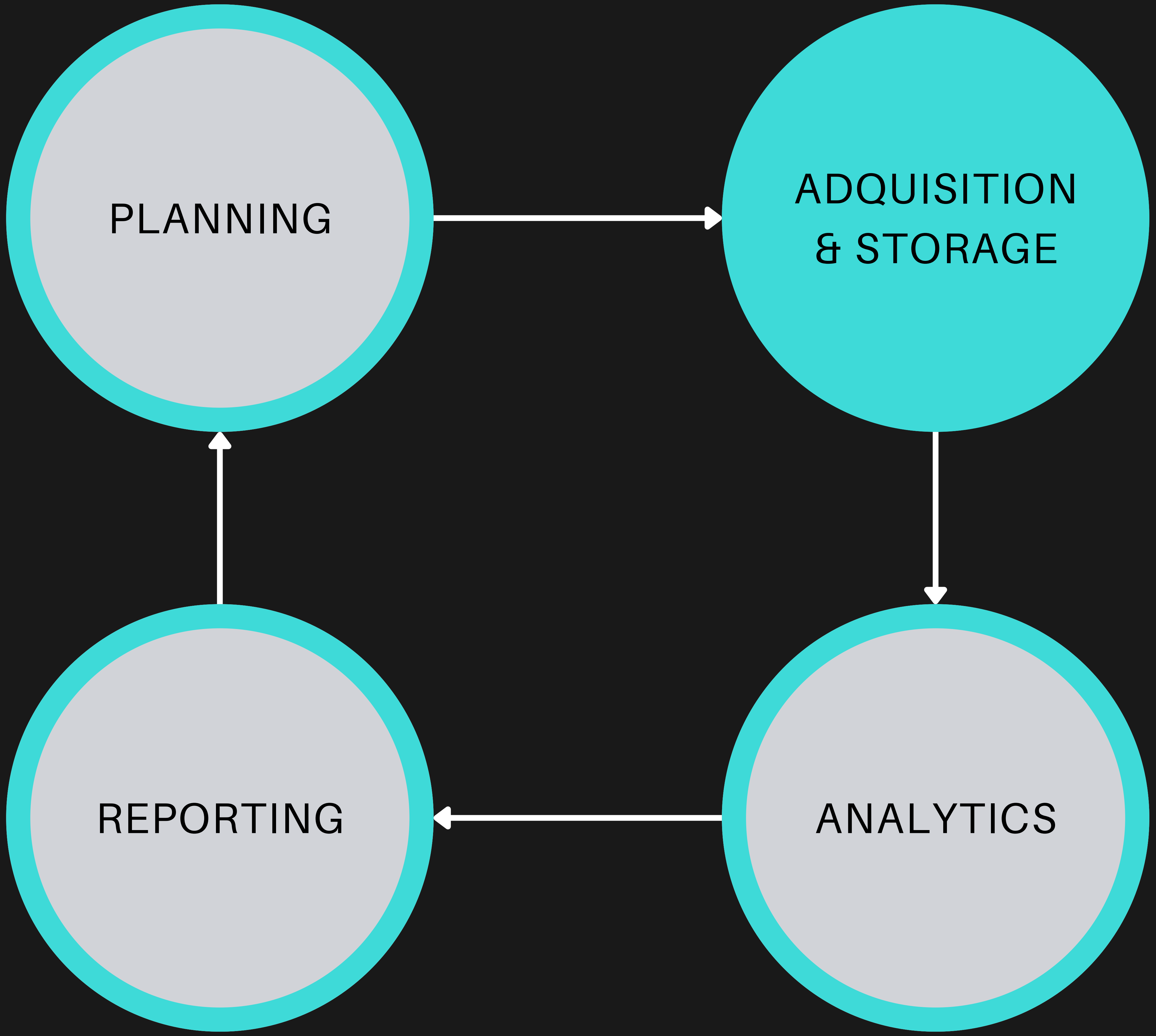
GAME ANALITICS

ADQUISITION & STORAGE



(My)SQL - FIRST CONTACT

1. Tables: We all know what tables are — a matrix of rows and columns. In databases, it’s the same. Each row is a record, or a unit of data. A record (row) can have several columns or fields. Each field is like an attribute of that record.
2. Queries: A query is a question posed to the database, to retrieve a specific set of records, based on conditions supplied in the query.
3. Views: These are virtual tables, or (a set of) stored queries.

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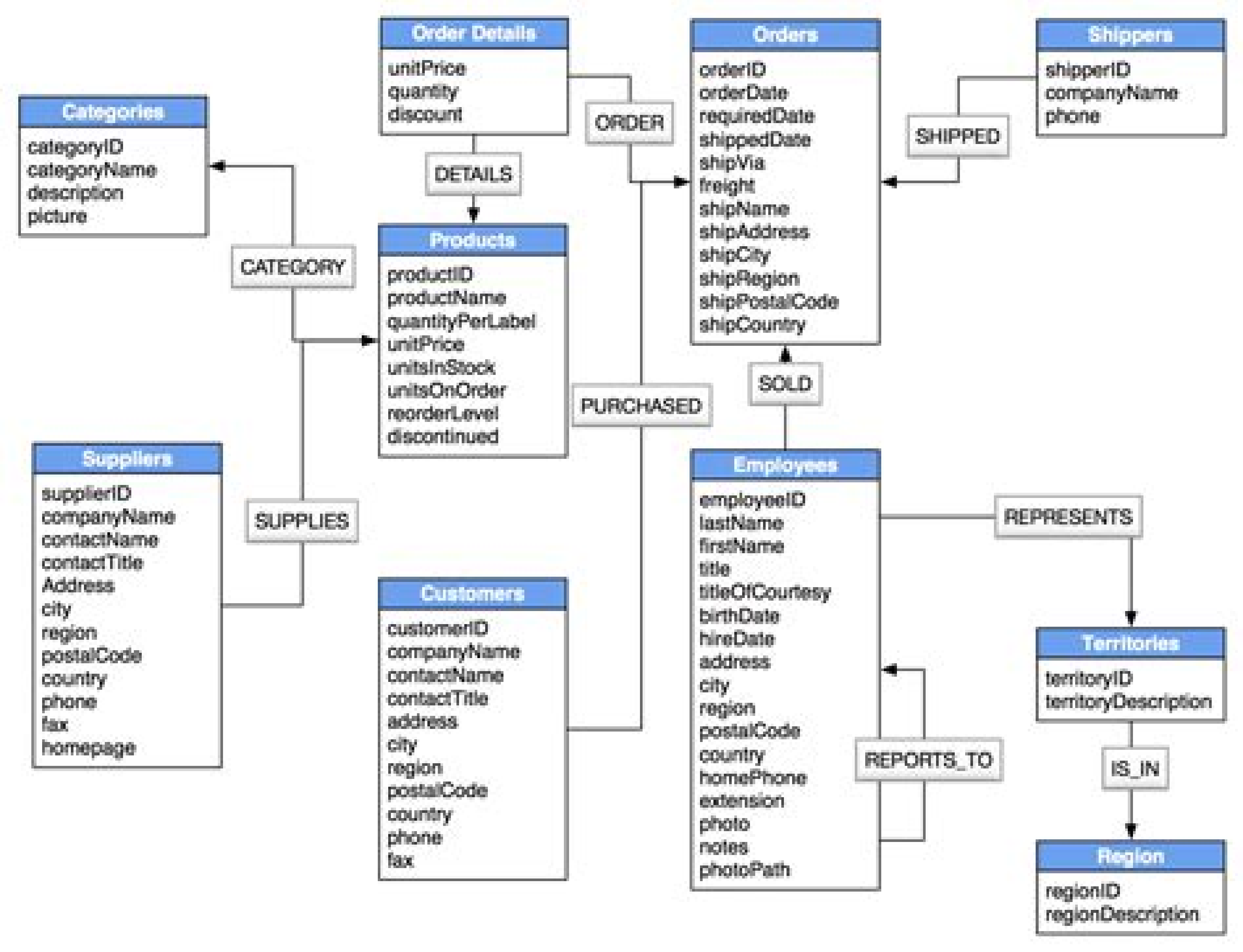
(My)SQL – RELATIONAL TABLES



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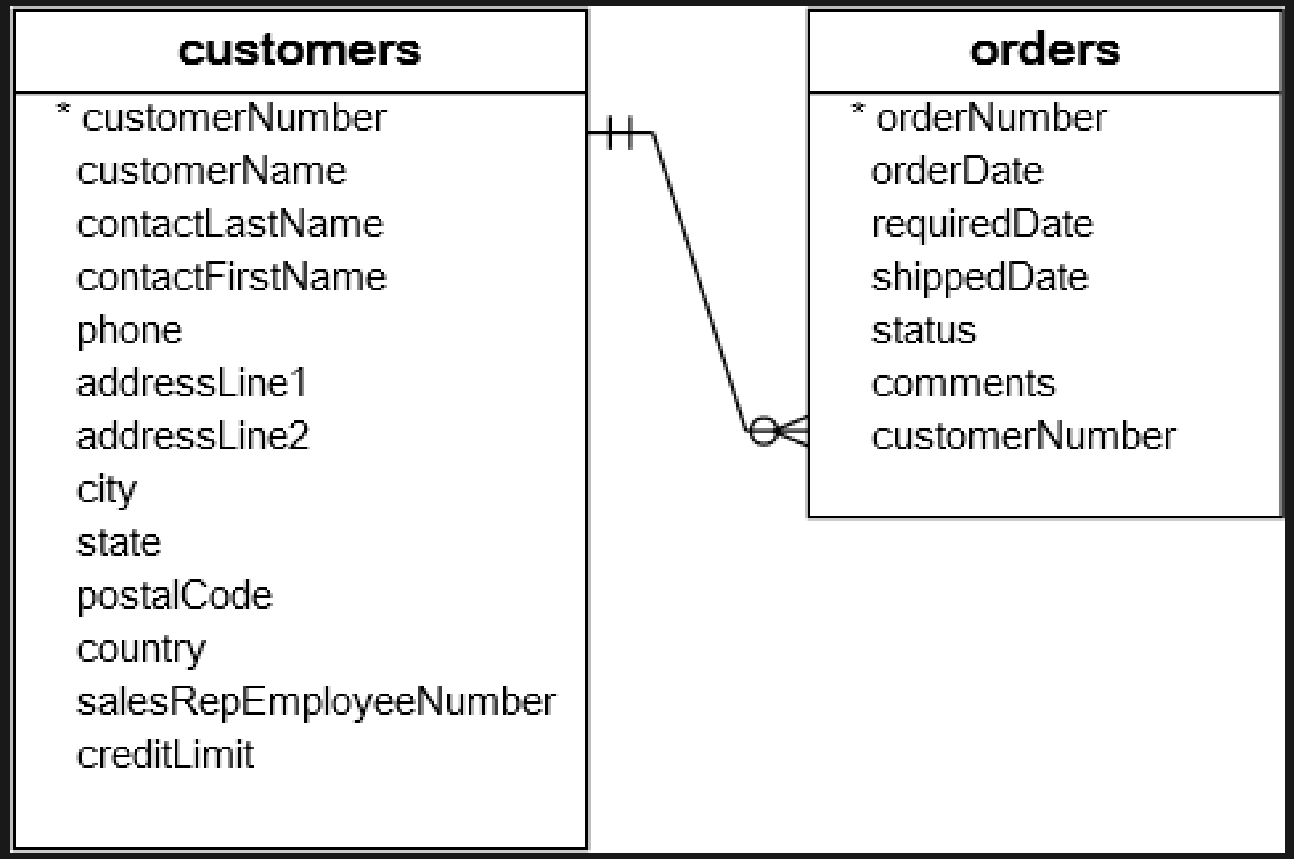
RELATIONAL TABLES - KEYS

A **primary key** is a column (or a set of columns) that uniquely identifies each row in the table. A primary key column cannot contain NULL values.

Foreign keys let you cross-reference related data across tables. Foreign key relationships involve a parent table that holds the central data values, and a child table with identical values pointing back to its parent. The FOREIGN KEY clause is specified in the child table.



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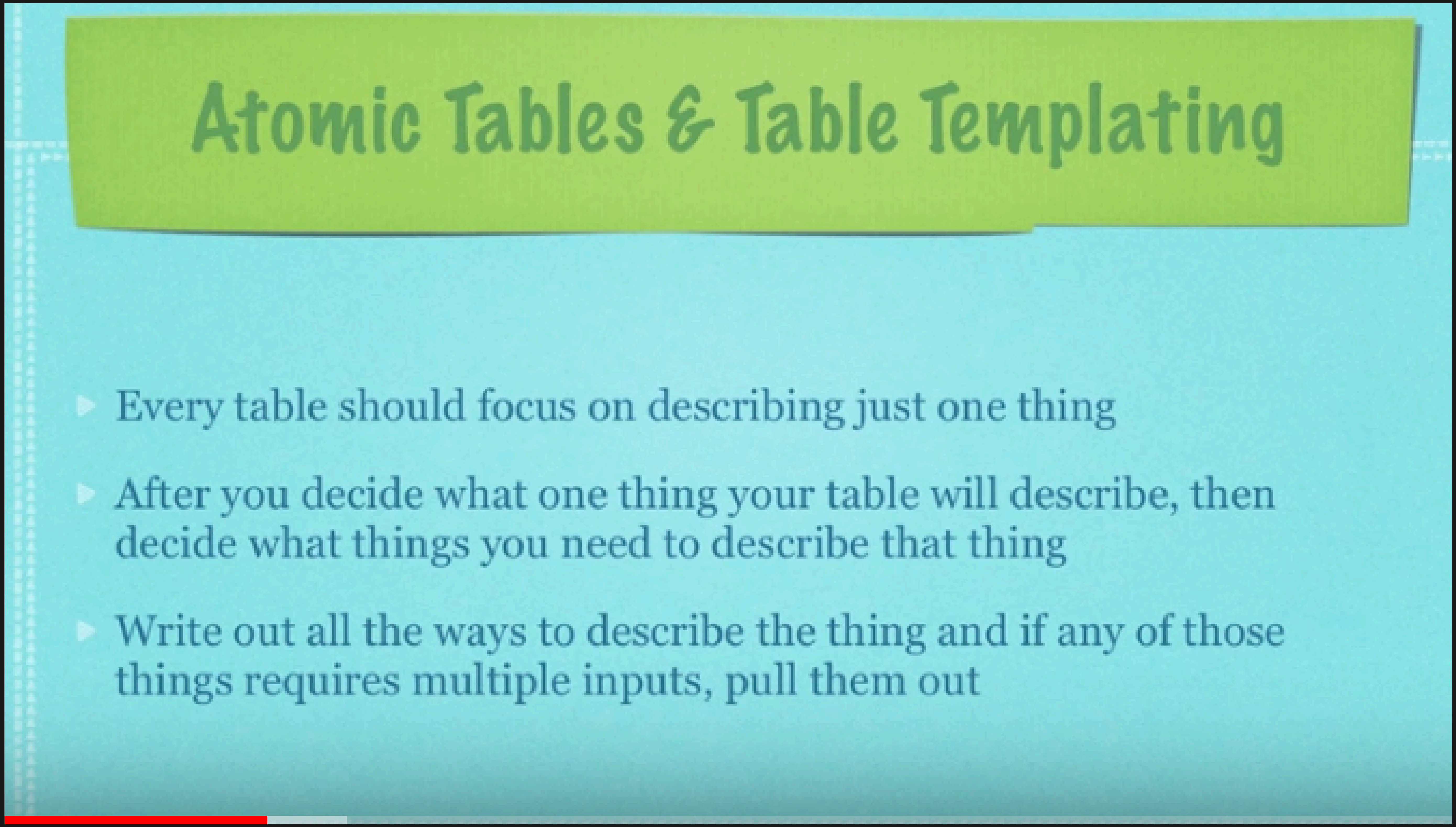

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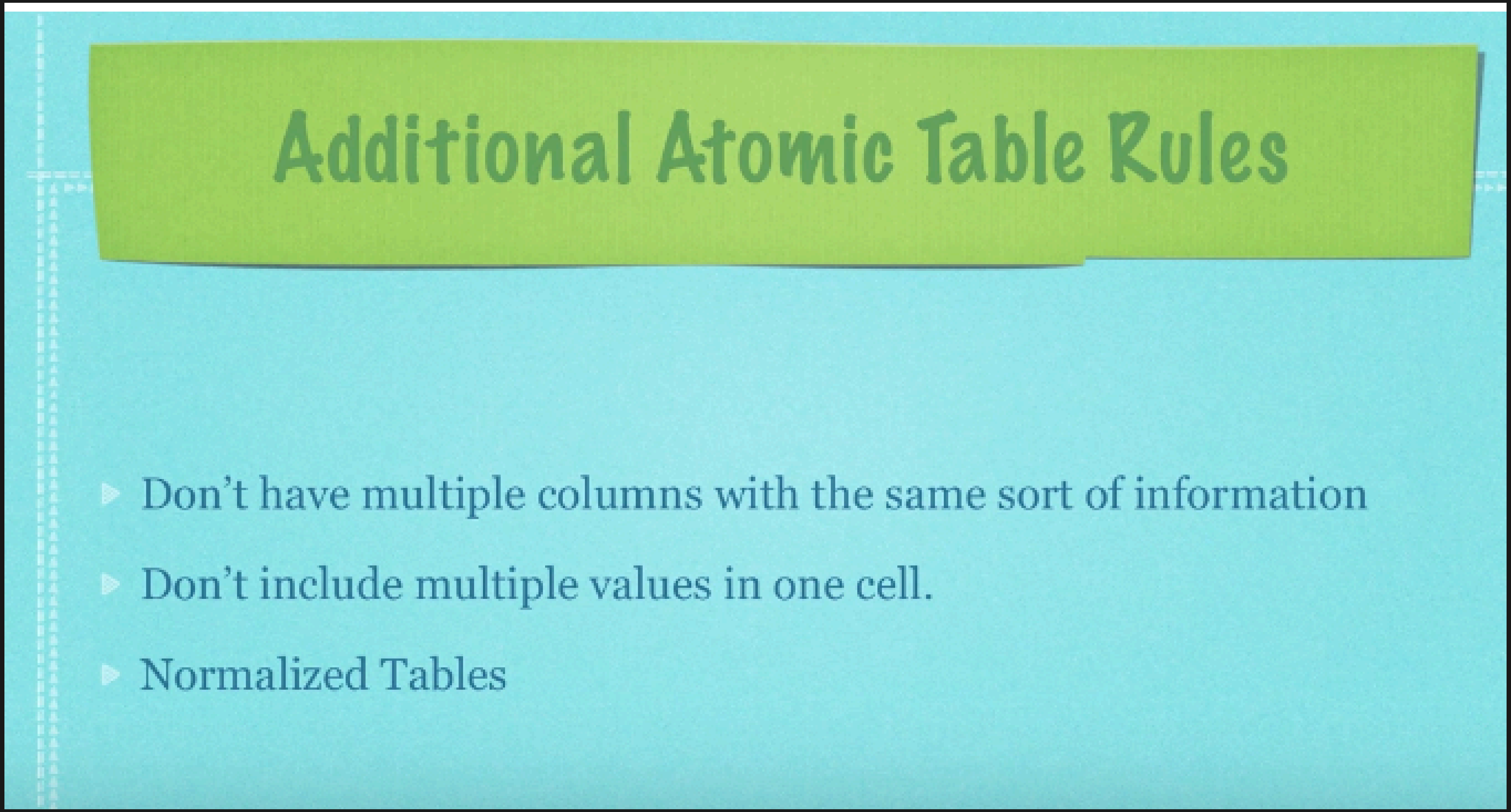

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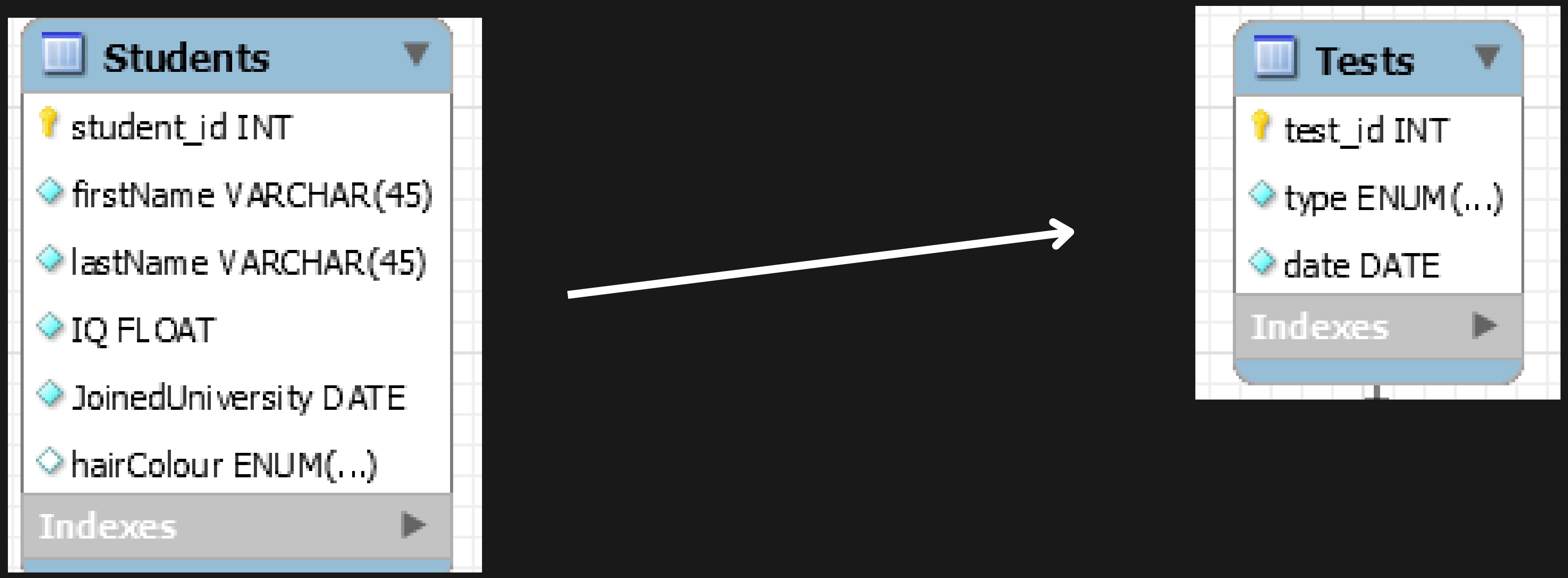

How can we store multiple test for multple students?

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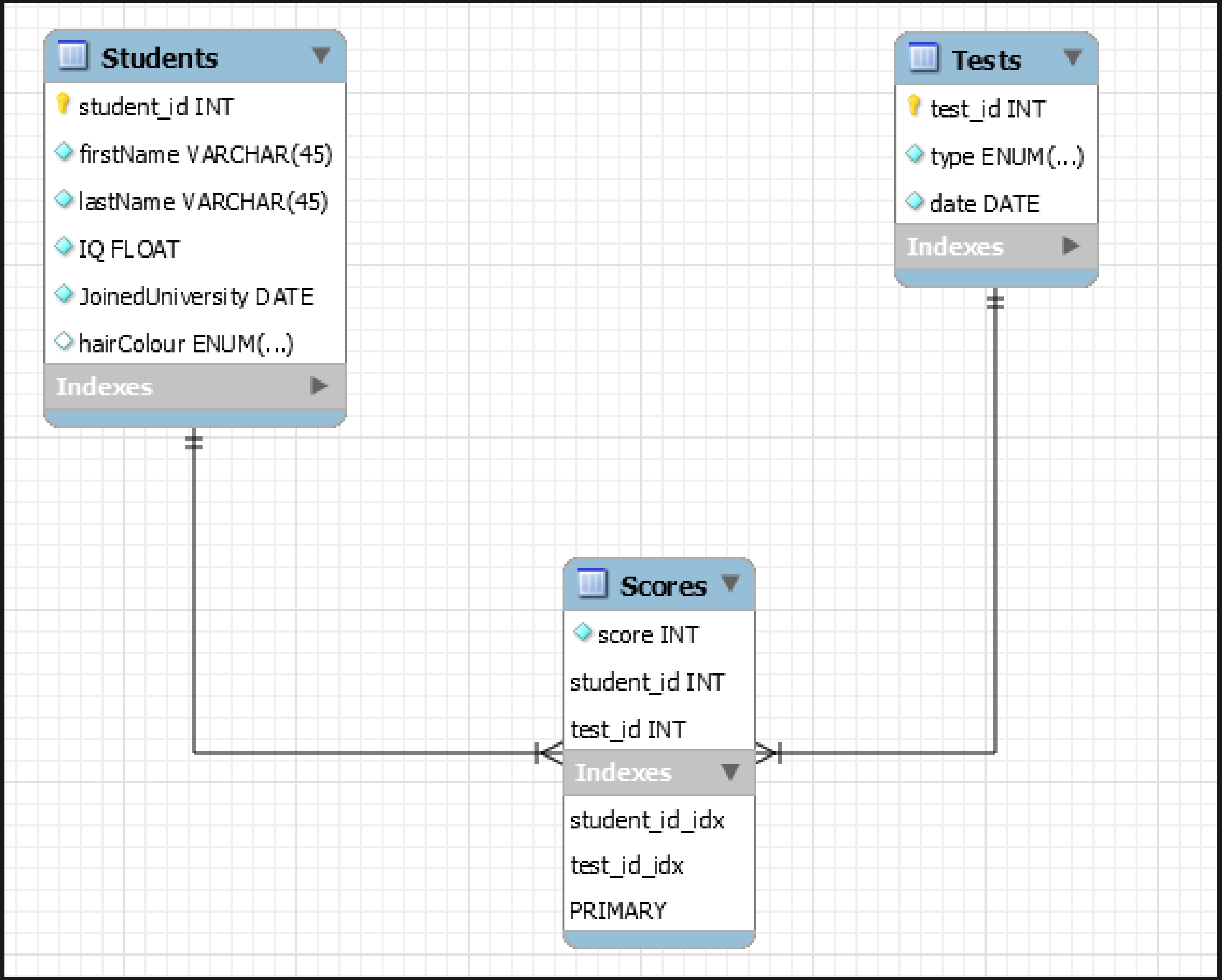
ADQUISITION & STORAGE

Add a third table that points to both.



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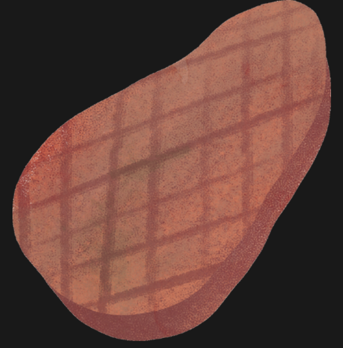


How should we store the data?

EventType	DayOfYear	DeviceID
StartSession	51	ohNtplp9qHT5
StartSession	81	5FLsykW7ZTZH
StartSession	62	iYZW9aMaHRP
StartSession	56	Oou6jpfmhW62
StartSession	71	CKeTdQMDS43
StartSession	36	VfQ3TimerKVI
StartSession	25	DRPUWFmHhi
StartSession	48	D8jscqjUidX8
StartSession	11	LKIDbUsuapLt
StartSession	79	Ph44PY8HdK4q
StartSession	55	ULE57L5d8Bz4
StartSession	70	iYZW9aMaHRP
StartSession	50	jekuazcAO0aX
StartSession	82	i3vYDAp0SkmF
StartSession	50	3b8X3tggleHZ
StartSession	59	CKeTdQMDS43
StartSession	44	m8XwRc7OONl
StartSession	15	sPVPpSKv6OcZ
StartSession	49	H4bflZx9Wfyg

DoY	DAU
1	0
2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	1
11	5
12	2
13	1
14	1
15	10
16	3
17	3
18	4
19	1
20	1
21	3
22	10

GAME ANALITICS

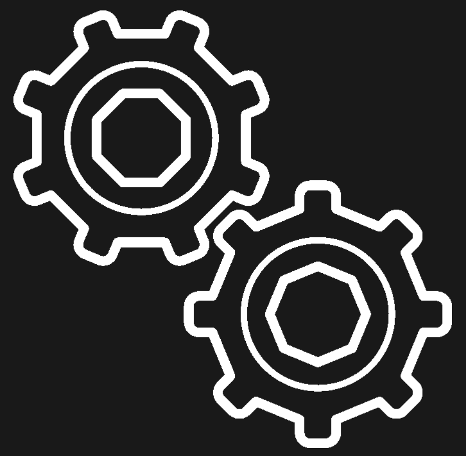
ADQUISITION & STORAGE

Number of users: DAU, MAU
Retention: Stickiness, D1,D3,D7
Monetisation: ARPU, ARPPU
Sessions: nº, length

(For all of them we want to know:
Age, Gender, Country)

Create all the tables needed to store the relevant info for those KPIs

10 min



GAME ANALITICS

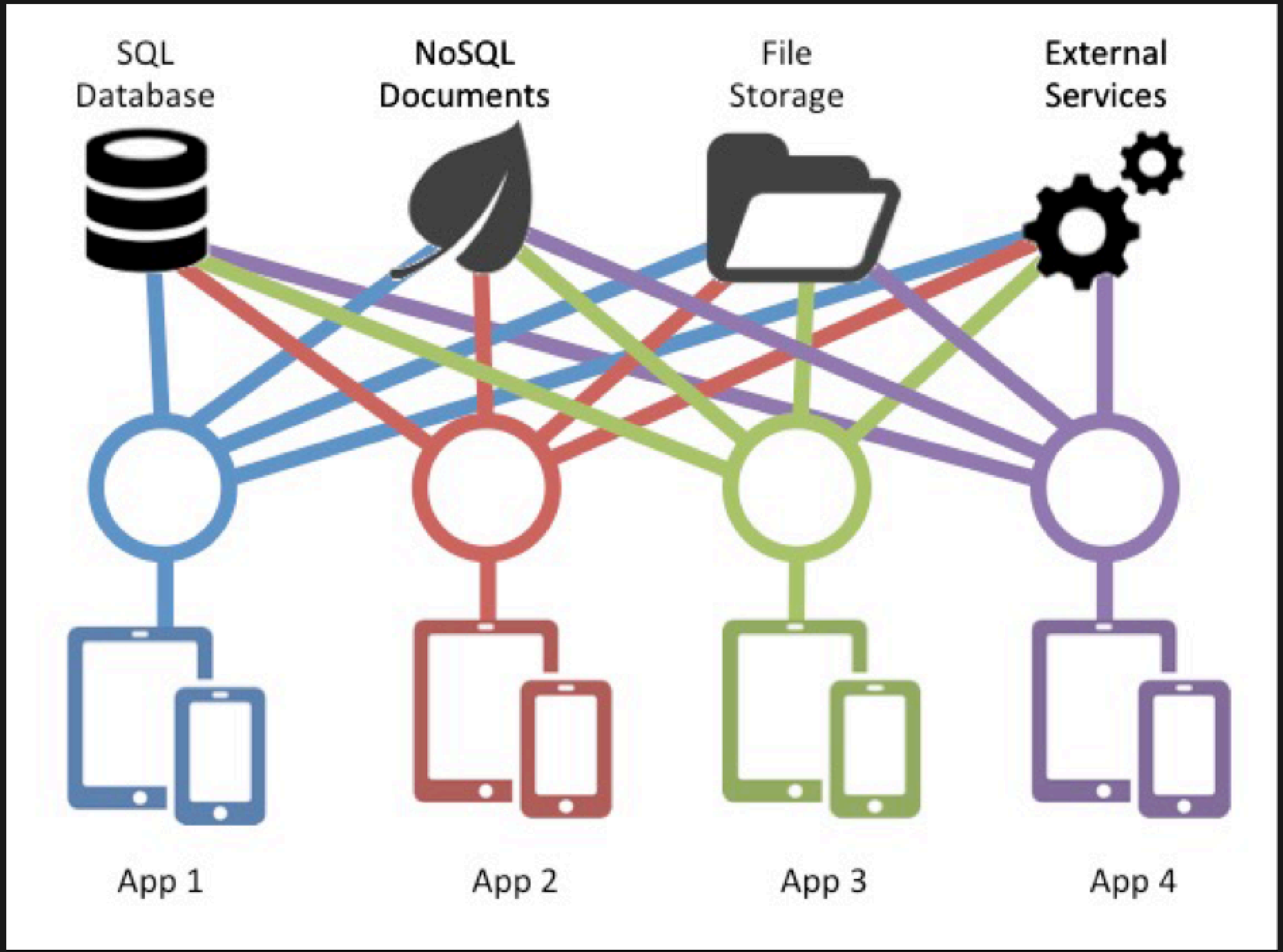
ADQUISITION & STORAGE

- SELECT
  - Retrieve data from one or more tables
- INSERT
  - Add new data to a table
- UPDATE
  - Change one or more rows in a table
- DELETE
  - Delete entries from a table

GAME ANALITICS

ADQUISITION & STORAGE

Server



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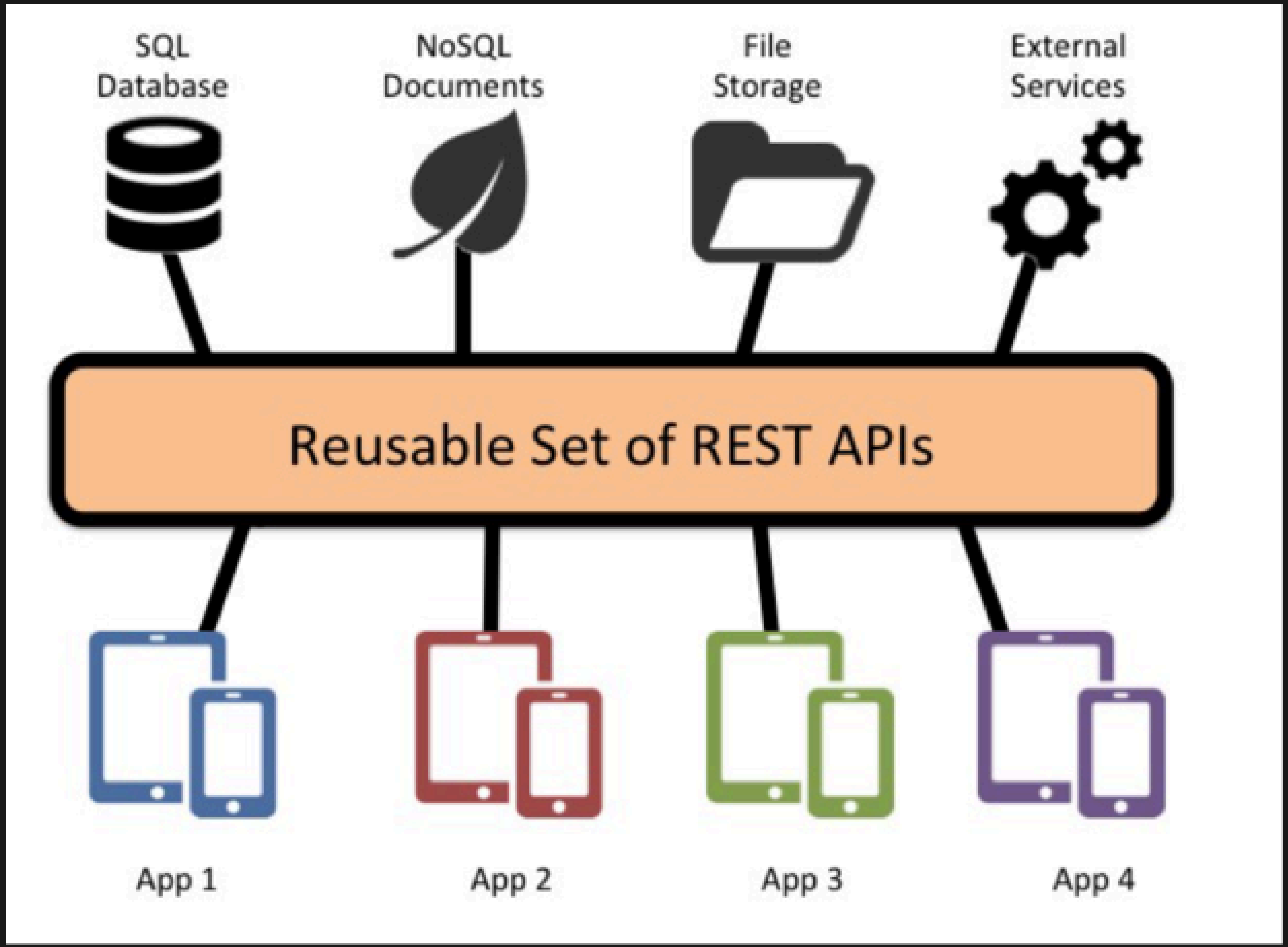
ADQUISITION & STORAGE

Server



GAME ANALITICS

ADQUISITION & STORAGE

Server - RESTful API

Client-server architecture: separation of concerns

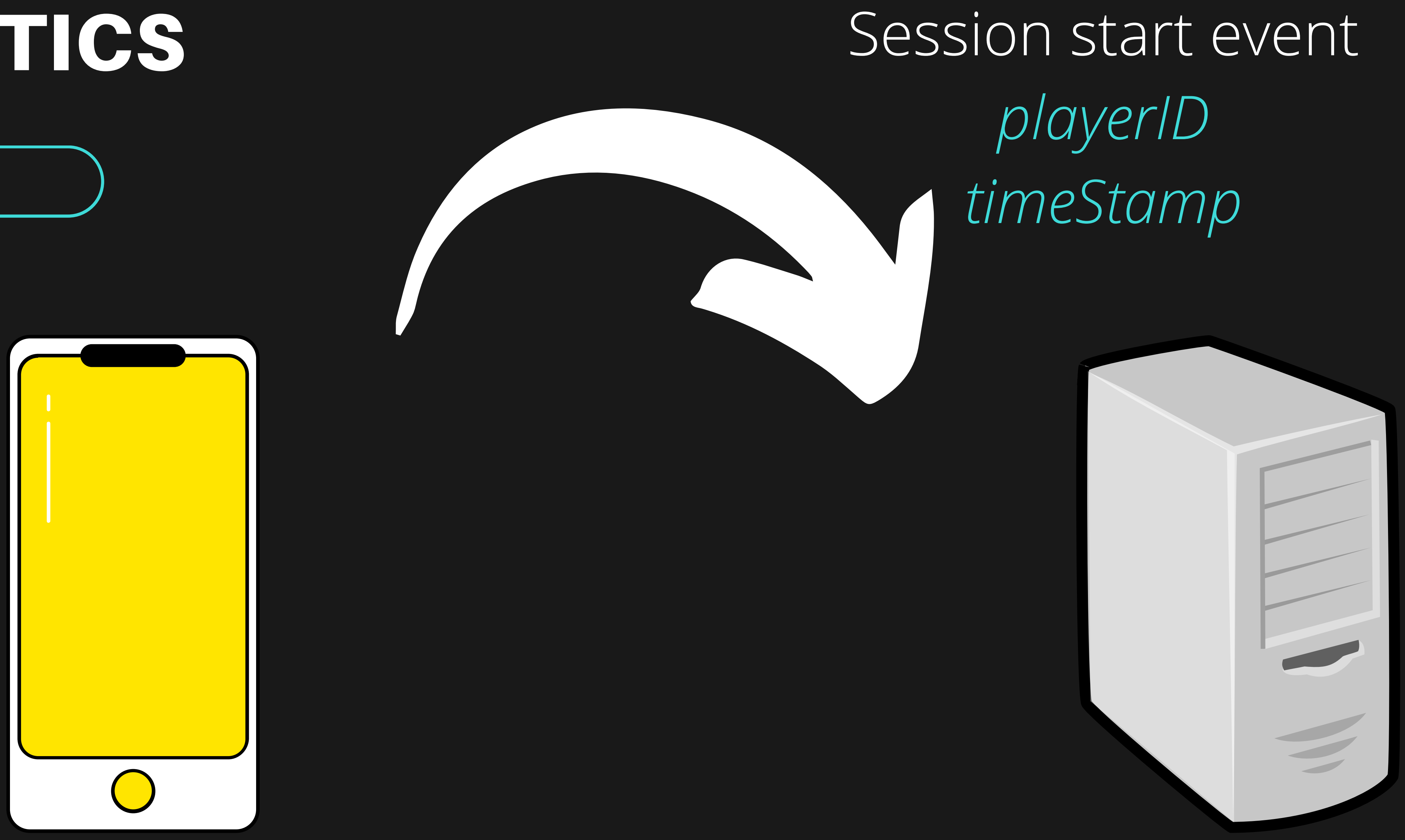
Statelessness: Each request from any client contains all the information necessary to service the request

Cacheability: clients and intermediaries can cache responses

Layered system: A client cannot ordinarily tell whether it is connected directly to the end server, or to an intermediary

GAME ANALITICS

ADQUISITION & STORAGE



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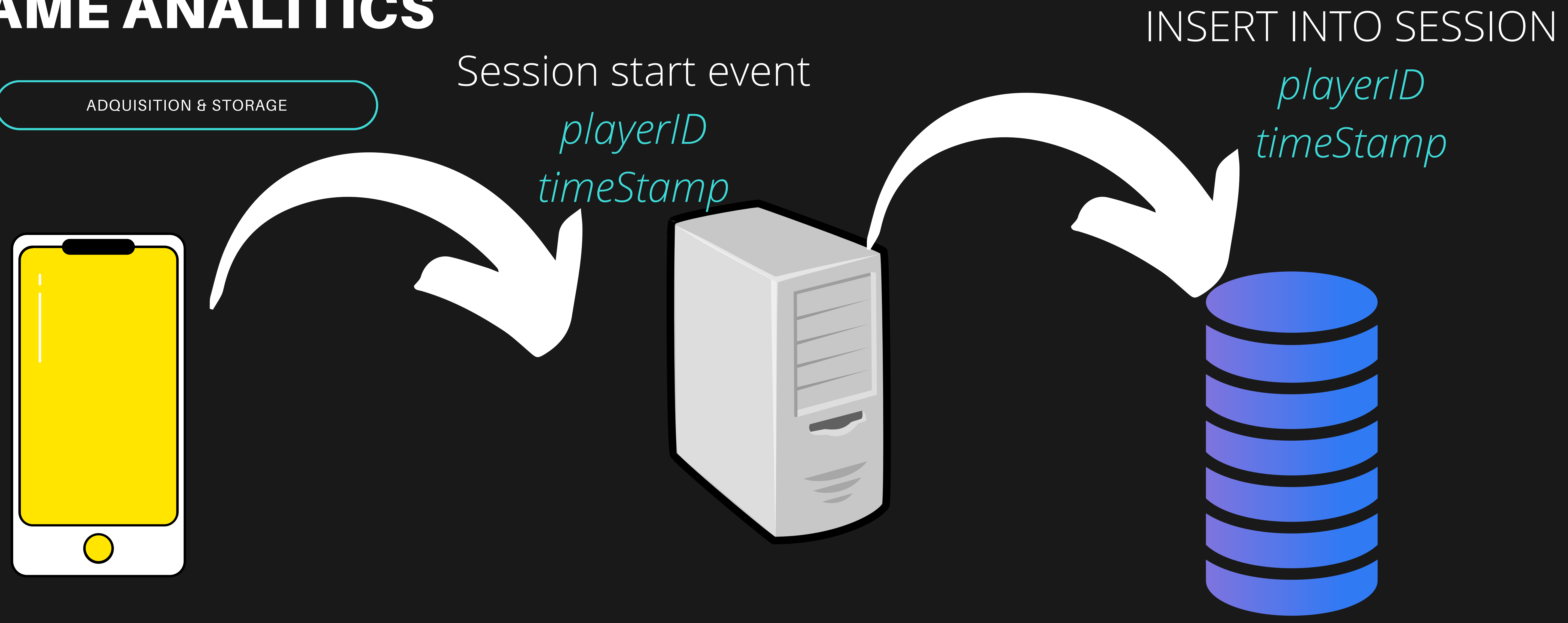

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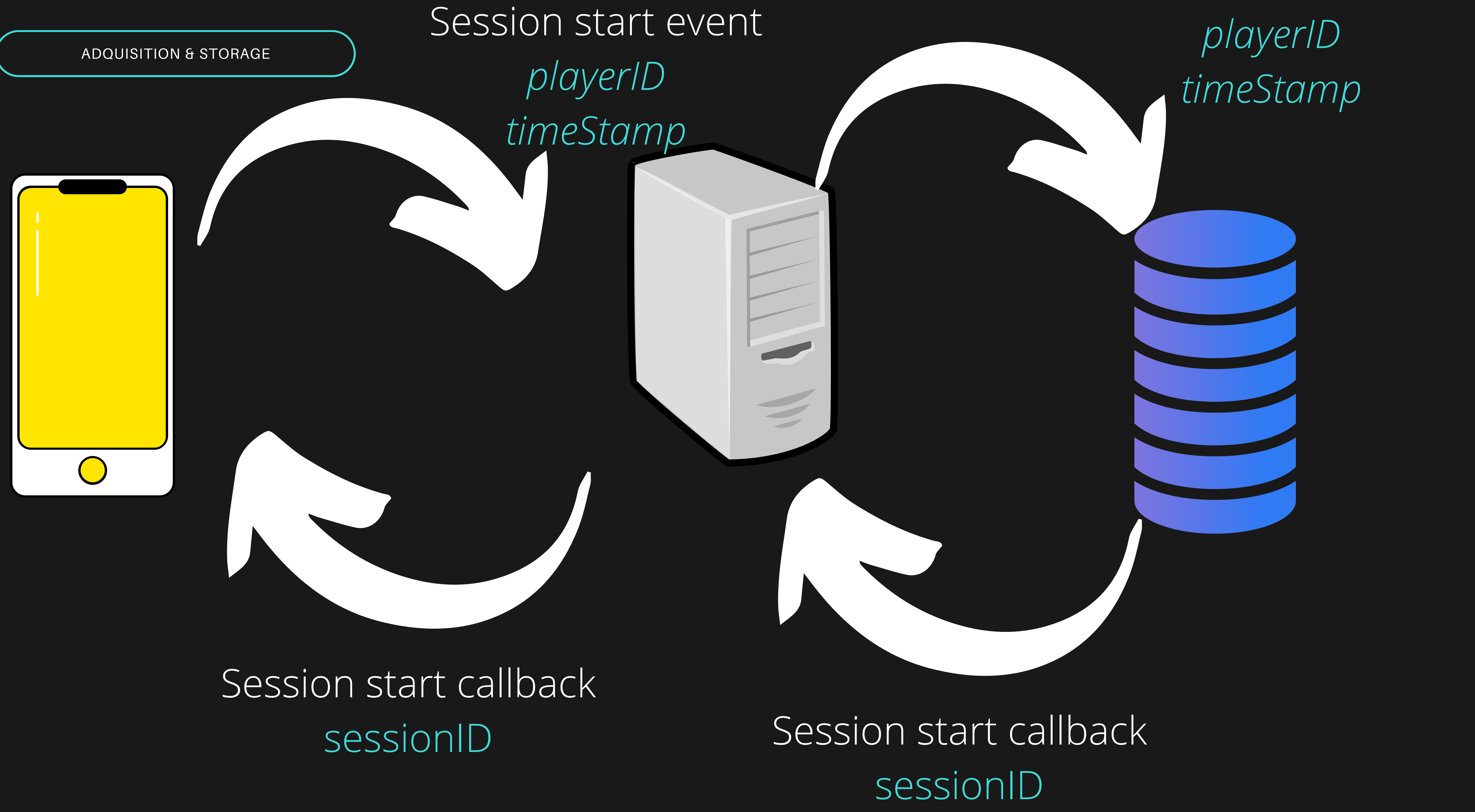

GAME ANALITICS

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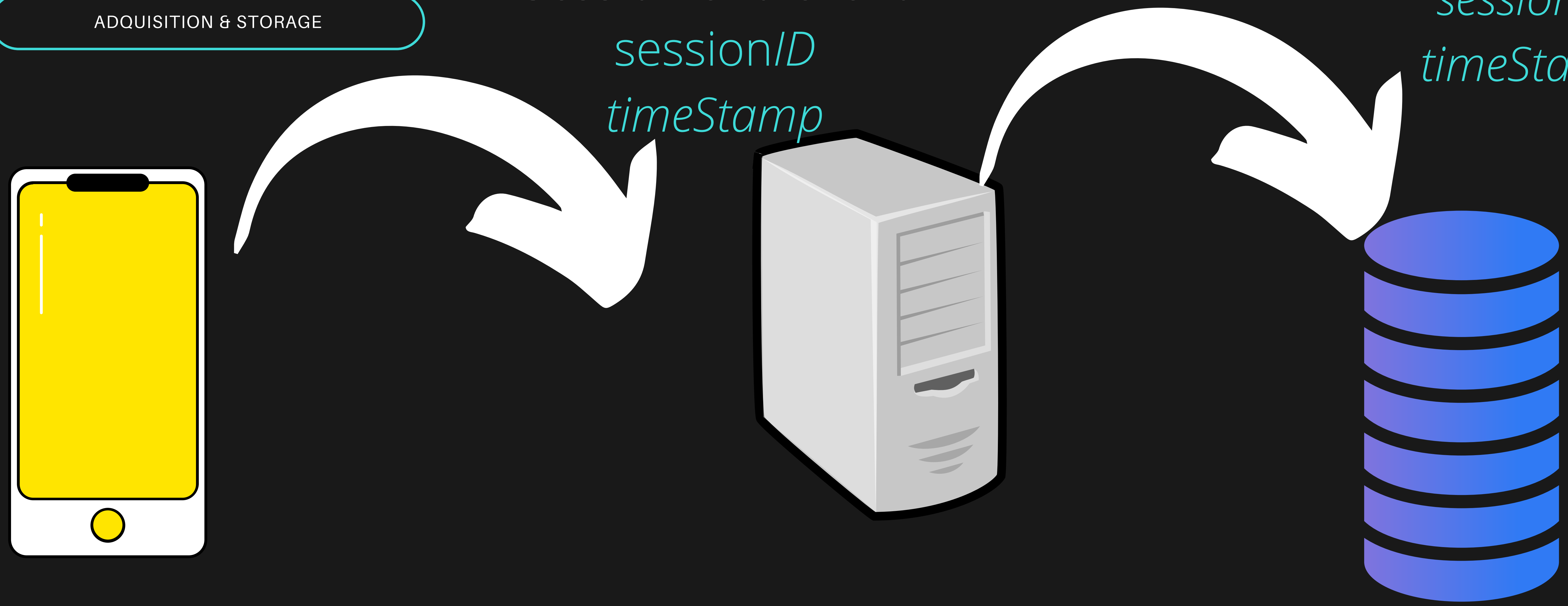

GAME ANALITICS

ADQUISITION & STORAGE

EventType	User	Date
StartSession	u1	13/03/2018 14:55:22
EndSession	u1	13/03/2018 14:58:42
StartSession	u1	27/03/2018 17:13:45
StartSession	u1	27/03/2018 19:51:02
EndSession	u1	27/03/2018 20:15:42

Unfinished session!!!

What can we do to prevent this?

GAME ANALITICS

ADQUISITION & STORAGE

Number of users: DAU, MAU
Retention: Stickiness, D1,D3,D7
Monetisation: ARPU, ARPPU
Sessions: nº, length

(For all of them we want to know:
Age, Gender, Country)

Write pseudo code for all needed events (on client and server)

10 min

SERIALISATION

Serialisation is the process of translating a data structure or object state into a format that can be stored (for example, in a file or memory data buffer) or transmitted (for example, over a computer network) and reconstructed later

SERIALISATION FORMAT

- CSV
- JSON
- Protobuf (Protocol Buffers)
- TOML
- XML
- YAML

```
id, name, age, address
4, Charmander, 12.34, Fire St
25, Pikachu, 56.78, Electric St
```

https://blog.mbedded.ninja/programming/serialization-formats/a-comparison-of-serialization-formats/

GAME ANALITICS

ADQUISITION & STORAGE

SERIALISATION FORMAT

CSV
JSON
Protobuf (Protocol Buffers)
TOML
XML
YAML


```json
[
    {
        "id": 4,
        "name": "Charmander",
        "age": 12.34,
        "address": "Fire Street"
    },
    {
        "id": 25,
        "name": "Pikachu",
        "age": 56.78,
        "address": "Electric Street"
    }
]
```


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```
#^R^Charmander^Z^Fire Street%<A2>@

^A^R^Pikachu^Z^Electric Street%<F1>'OA
```


https://blog.mbedded.ninja/programming/serialization-formats/a-comparison-of-serialization-formats/

ADQUISITION & STORAGE

SERIALISATION FORMAT

CSV
JSON
Protobuf (Protocol Buffers)
TOML
XML
YAML

```toml
[[pokemon]]
id = 4
name = "Charmander"
address = "Fire Street"
age = 12.34

[[pokemon]]
id = 25
name = "Pikachu"
address = "Electric Street"
age = 56.78
```

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ADQUISITION & STORAGE

SERIALISATION FORMAT

CSV
JSON
Protobuf (Protocol Buffers)
TOML
XML
YAML

```xml
<people>
    <person>
        <id>4</id>
        <name>Charmander</name>
        <age>12.34</age>
        <address>Fire St</address>
    </person>
    <person>
        <id>25</id>
        <name>Pikachu</name>
        <age>56.78</age>
        <address>Electric Street</address>
    </person>
</people>
```

https://blog.mbedded.ninja/programming/serialization-formats/a-comparison-of-serialization-formats/

SERIALISATION FORMAT

CSV
JSON
Protobuf (Protocol Buffers)
TOML
XML
YAML

```yaml
- { id: 0, name: Charmander, age: 12.34, address: "Fire Street" }
- { id: 25, name: Pikachu, age: 56.78, address: "Electric Street" }
```

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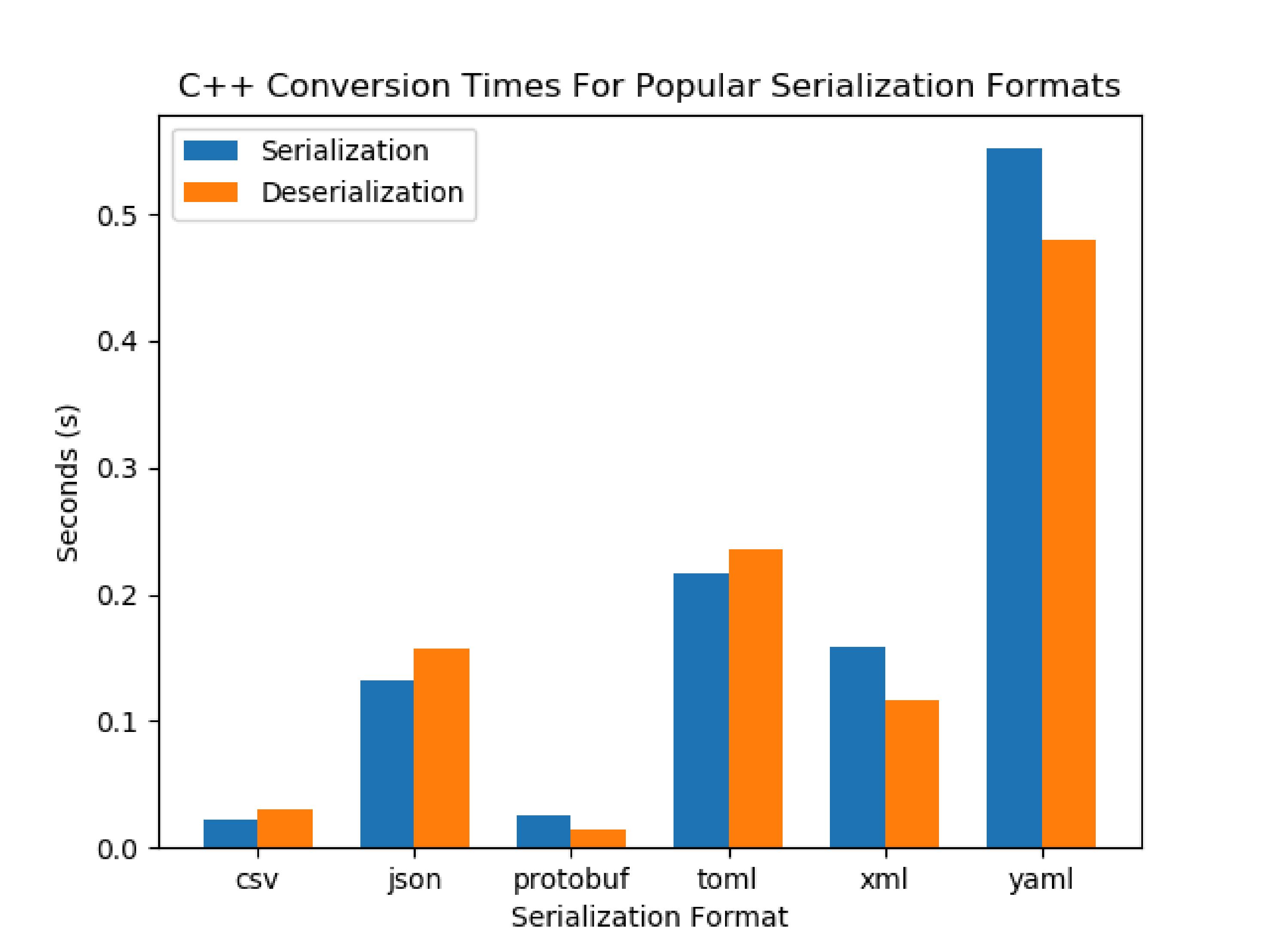

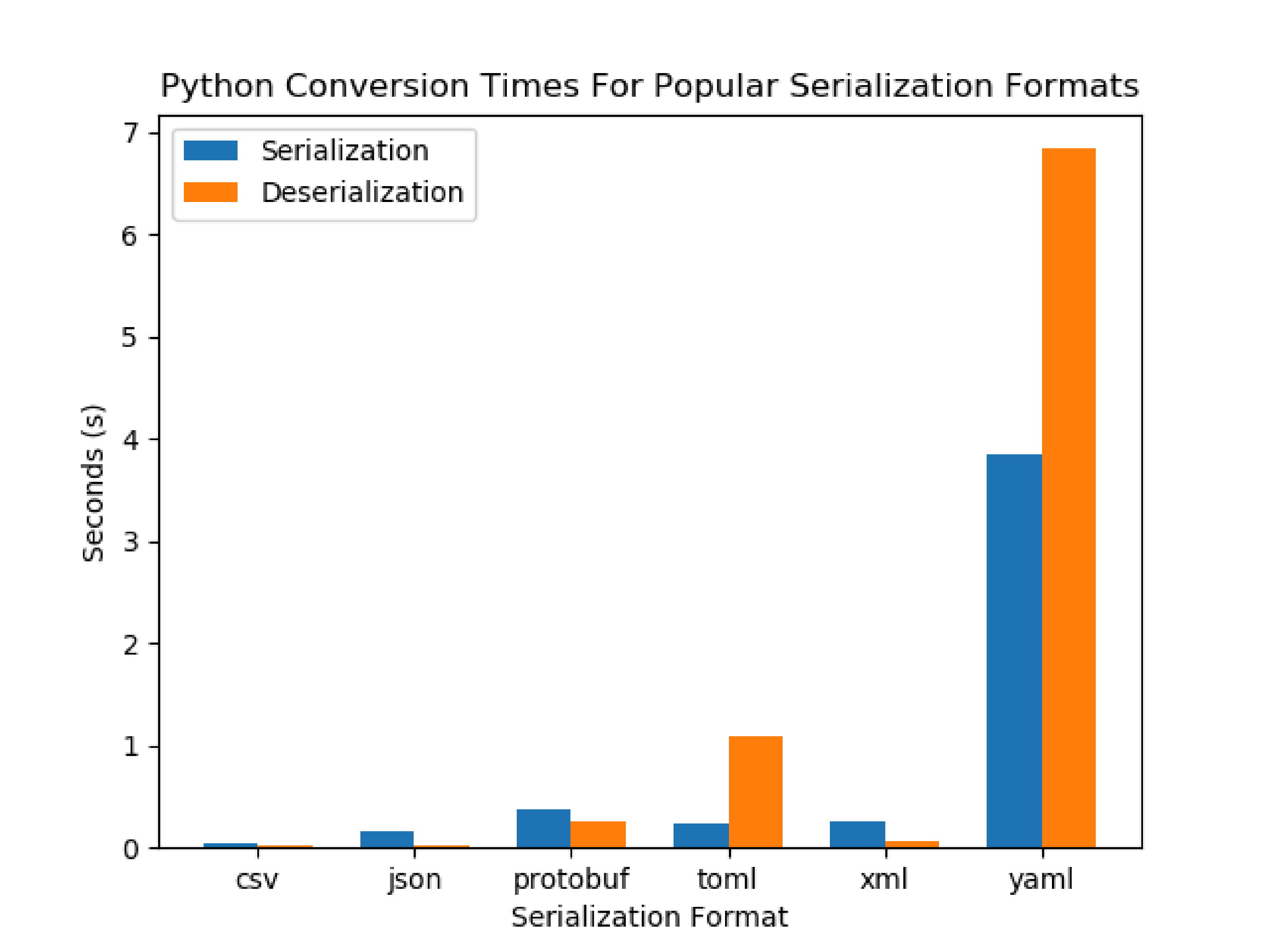

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Relevant Questions

Know what the mentioned KPIs mean

Know how good a KPI is (benchmark)